

Figure 2: Universal scaling regimes for causal synchronization

Code description for GitHub repository (Mevan Rajakaruna)

M. Rajakaruna

Overview

This Python script generates Figure 2 of the manuscript entitled “Feedback polarity controls causal clustering and cascade propensity in stochastic rupture”. The figure comprises four panels (a–d) that illustrate the scaling regimes for causal synchronization under displacement control, the exponential suppression under force control, a phase diagram relating feedback polarity to loading rate, and the critical pulling rate required for cascading failure. The script includes analytical curve plotting, synthetic Monte Carlo data points with error bars, and deterministic phase boundary calculations. No experimental data are used; all simulation points are generated synthetically for visual illustration.

1 Software and libraries

The script requires Python version 3.9 or higher. Numerical array operations are handled by NumPy (version 1.21 or later). Statistical computations and curve fitting functions are imported from `scipy.stats` and `scipy.optimize`, though the latter is not actively used in the current version. All figure construction, including subplot layout, patches, and file saving, is performed using Matplotlib (version 3.4 or later) with the `gridspec` module for custom subplot arrangements.

2 Plotting style

Publication-quality visual styling is enforced through customisation of `matplotlib.rcParams`. The font family is set to sans-serif (Arial or Helvetica), with a base font size of 9 points. Axes labels are 10 points, titles 10 points, tick labels 8 points, and legend text 8 points. The figure is rendered at 300 DPI for screen display and saved at 600 DPI. Line widths are configured as follows: axes lines 0.8 points, tick marks 0.8 points, and main curves 1.5 points. The bounding box is tight with a padding of 0.05 inches.

3 Figure layout

A 3×2 grid is created using `gridspec.GridSpec` with height ratios $[1, 1, 0.8]$, horizontal spacing of 0.45, and vertical spacing of 0.30. Panel a occupies the entire first row. Panel

b occupies the left column of the second row. Panel c occupies the right column of the second row. Panel d occupies the entire third row.

4 Panel a: Displacement control scaling

Panel a demonstrates the linear scaling of the causal synchronization probability P_c with the dimensionless parameter $\alpha k v t_{\text{mat}}$ under displacement control. The analytical expression implemented is $P_c = (r_0 e^{-\alpha k \ell_0} + \frac{\alpha k v}{2}) t_{\text{mat}}$, normalised to the form $P_c = (r_0^{\text{norm}} + x/2)$ where $x = \alpha k v t_{\text{mat}}$ and $r_0^{\text{norm}} = r_0 e^{-\alpha k \ell_0} t_{\text{mat}}$.

Three thermal contribution values are plotted: $r_0^{\text{norm}} = 0.01$ (blue), 0.1 (orange), and 0.5 (green). The loading-dominated asymptotic regime $P_c \sim (\alpha k v / 2) t_{\text{mat}}$ is shown as a black dashed line. Synthetic Monte Carlo simulation points are generated for demonstration purposes at $x = 0.1, 0.3, 0.5, 0.8, 1.2, 1.6$. For each point, the mean is computed as $P_c = x/2 + 0.05 \cdot \mathcal{N}(0, 1)$ where $\mathcal{N}(0, 1)$ is a standard normal random variate, with a fixed random seed of 42 for reproducibility. Error bars are set to a constant value of ± 0.02 . The thermal regime ($x < 0.3$) is shaded in light gray, and the loading-dominated regime ($x > 0.3$) is shaded in light coral. The axis limits are $x \in [0, 2]$ and $P_c \in [0, 1.2]$. The legend is placed in the upper left corner with no frame and full transparency.

5 Panel b: Force control exponential suppression

Panel b illustrates the exponential suppression of causal synchronization under force control with transient unloading. The suppression factor is given by $P_c / (r_0 e^{\alpha F} t_{\text{mat}}) = e^{-\alpha F \beta}$, where $\beta \in [0, 1]$ is the unloading fraction. Three values of the load sensitivity parameter are plotted: $\alpha F = 1$ (red), 2 (purple), and 5 (brown). The curve is computed as $\exp(-\alpha F \cdot \beta)$ over 1000 linearly spaced β values. A vertical dashed black line at $\beta = 0$ is annotated as the ideal force clamp limit. A light yellow box embedded in the plot highlights the $e^{-\alpha F \beta}$ suppression mechanism. The axis limits are $\beta \in [0, 1]$ and the normalised probability $P_c / (r_0 e^{\alpha F} t_{\text{mat}}) \in [0, 1.05]$. The legend is placed in the upper right corner with no frame and full transparency.

6 Panel c: Phase diagram

Panel c presents a schematic phase diagram relating feedback polarity to the loading rate parameter $\alpha k v t_{\text{mat}}$. A phase boundary is placed at the microscopic onset criterion $P_c = \epsilon$ with $\epsilon = 0.1$. Solving $P_c = \epsilon$ under loading-dominated conditions yields the critical condition $\alpha k v t_{\text{mat}} = 2\epsilon = 0.2$. The positive feedback (displacement control) region is shown as a horizontal light coral bar spanning $\alpha k v t_{\text{mat}} \in [0, 3]$ at vertical position $y = 0$, representing the active collective phase where cascading failure is possible. The negative feedback (force control) region is shown as a light blue bar at vertical position $y = -0.5$, representing the absorbing phase where failure remains sequential and localised. A vertical red line of linewidth 2 marks the phase boundary at $x = 0.2$. The x-axis is labeled “Loading rate $\alpha k v t_{\text{mat}}$ ” and the y-axis is labeled “Feedback polarity” with tick labels “Positive” at $y = 0$ and “Negative” at $y = -0.5$. The axis limits are $x \in [0, 3]$ and $y \in [-1, 0.5]$. The legend is placed in the lower right corner with no frame and full transparency.

7 Panel d: Critical pulling rate for cascading failure

Panel d shows the critical pulling rate required to initiate cascading failure as a function of the material time $t_{\text{mat}} = L/c_s$. From the onset criterion $\alpha k v t_{\text{mat}} = 2\epsilon$ with $\epsilon = 0.1$, the critical pulling rate is derived as $v_c = 2\epsilon/(\alpha k t_{\text{mat}})$. For visualisation, αk is set to unity, giving $v_c = 0.2/t_{\text{mat}}$. The curve is plotted in red over $t_{\text{mat}} \in [0.01, 1]$ with 100 linearly spaced points. The region above the curve ($v > v_c$) is filled in light coral and labeled “Cascading failure”. The region below the curve ($v < v_c$) is filled in light blue and labeled “Sequential failure”. The axis limits are $t_{\text{mat}} \in [0, 1]$ and $v_c \in [0, 20]$ in normalised units. The legend is placed in the upper right corner with no frame and full transparency.

8 Output files

The script saves the figure in three formats. An Encapsulated PostScript file is written as `fig2_scaling.eps`. A high-resolution PNG file at 600 DPI is written as `fig2_scaling.png`. A Portable Document Format file is written as `fig2_scaling.pdf`. All files are saved in the current working directory. The figure is also displayed interactively via `plt.show()`, and the script prints a confirmation message upon completion.

9 Reproducibility

The random number generator is initialised with seed 42 using `np.random.seed(42)` to ensure reproducible synthetic Monte Carlo points. No external data files are required. The synthetic Monte Carlo points are generated solely for visual illustration and do not correspond to actual simulation output from the stochastic rupture model; they follow the analytical trend with added Gaussian noise of standard deviation 0.05. To regenerate the figure exactly, the script must be executed in a Python environment with the specified library versions.

10 Relation to the manuscript

Panel a corresponds to Figure 2a in the paper, showing the scaling regimes for displacement control including the crossover from thermally dominated to loading-dominated behaviour. Panel b corresponds to Figure 2b, demonstrating the exponential suppression factor $e^{-\alpha F\beta}$ under force control. Panel c corresponds to Figure 2c, presenting the schematic phase diagram with the phase boundary at $\alpha k v t_{\text{mat}} = 2\epsilon$. Panel d presents supplemental information showing the critical pulling rate v_c as a function of the material time t_{mat} , derived from the onset criterion.

License and citation

This code is part of the research by M. Rajakaruna and H. Rajakaruna. If you use this code or the generated figure, please cite the associated manuscript as described in the `README.md` file in the repository.